MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (COMPUTER SCIENCE AND ENGINEERING)

PROCTORED ONLINE EXAMINATIONS, DECEMBER 2021

SUBJECT: MULTIMEDIA TECHNOLOGIES (PROGRAM ELECTIVE-V)

[CSE 4072]

REVISED CREDIT SYSTEM
(17/12/2021)

Time: 75 Mins MAX. MARKS: 20

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data, if any, may be suitably assumed.
- Write the answers on sheets of paper preferably on one side, scan all the sheets into a single pdf file, upload, Turn in /Hand in.
- Write your name, Registration number and put your signature on all the sheets

PART B

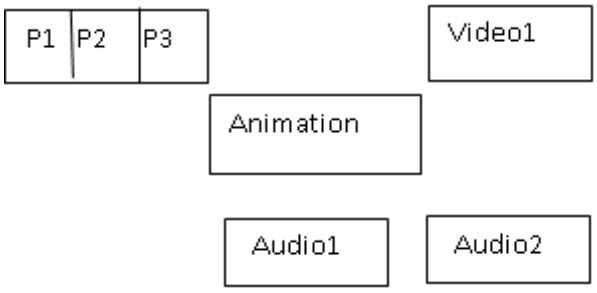
Q.No	Question statement	Marks
1.A	(i)Derive the bit rate and the memory requirements to store each frame that results from the digitization a 16CIF video format. Also find the total memory required to store a 1.5 hour conference video. Show all the steps clearly. The resolution of 16 CIF is Y=1440 X 1152 Cb=Cr =720X 576 Line sampling rate is 13.5 MHz for Y component and 6.75 for Cb and Cr components, all with 8 bits per sample (ii)Given a set of DCT coefficients and the quantization table as shown in Fig.Q 1.A, a) Determine the result of the quantization step of the JPEG compression method, applied to the block given in Fig. Q 1 A (i) , using the quantization table given in Fig. Q 1.A.(ii). b) Calculate the quantization error for each of the DCT coefficients. c) Determine the result of the zig-zag step being applied to the quantized block. d) Determine the result of the run length encoding (RLE) step being applied to the zig-zag step's output.	2 3

256	120	90	70	10	5	0	0
140	100	75	10	4	2	0	0
100	208	144	128	96	0	0	0
80	10	5	4	2	0	0	0
12	5	0	0	0	0	0	0
5	3	0	0	0	0	0	0
2	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Fig. Q.1A (i)DCT coefficients

10	10	15	20	25	30	35	40
10	15	20	25	30	35	40	45
15	20	25	30	35	40	50	60
20	25	30	35	40	50	60	70
25	30	35	40	50	60	70	80
30	35	40	50	60	70	80	90
35	40	50	60	70	80	90	100
40	50	60	70	80	90	100	110

(ii) Quantization table

1.B	Assume that the characters occur with probabilities of M = 0.1, E = 0.3,D = 0.3, I = 0.2 and A = 0.1. Using Arithmetic coding, encode the following stream of characters MEDIA Show clearly all the steps	3
1.C	Assume 2 X 2 macroblock is used for P frame encoding. For the following macroblock # # # # # 5 3 # # 2 3 # # # # # the corresponding intensities in the reference frame are given as follows: 5 3 6 2 1 4 7 2 4 5 3 3 3 2 3 2 Calculate the motion vector, with complete search within ±1 pixel search window. Indicate all the steps.	2
2.A	Define synchronization specification. For the application involving the media objects as shown in Fig.Q 2.A, give the synchronization specification using the following synchronization models. Also explain how you have arrived at each of the specification model that you have written. (i) Interval Based (ii) Hierarchical model (iii) Specification based on Global Timer  Fig.Q 2.A	5

2.B	Name the transport layer protocol used in Real Time Transport Protocol and list out the reasons for using it. Also describe how RTP creates its own timing and synchronization mechanism in order to ensure ordering	3
2.C	Define CDN? Discuss the similarities and differences between proxy caching and CDN. Is it beneficial to utilize both of them in a system? Justify your answer	2